

Impact of digitalization on power consumption

The aim of this analysis is to identify the most vulnerable EU countries, whose power consumption may be influenced by digitalization. Based on raw data from Eurostat, IEA and ENTSO-E webpages, each EU country is examined from three different perspectives:

- how much its power consumption is related to data centers and IT demands
- how fast its economy is moving towards digitalization
- how unstable its power grid has been in the recent years.

The analysis covers three reference years (2026, 2028, 2030), to show a trajectory rather than a static snapshot.

The final outcome is a matrix where all these KPIs are layered to find out which countries are more likely to face great changes due to IT developments.

Notes

I conducted this analysis in 2026 as an attachment to my application for the Senior Data Officer vacancy at IEA.

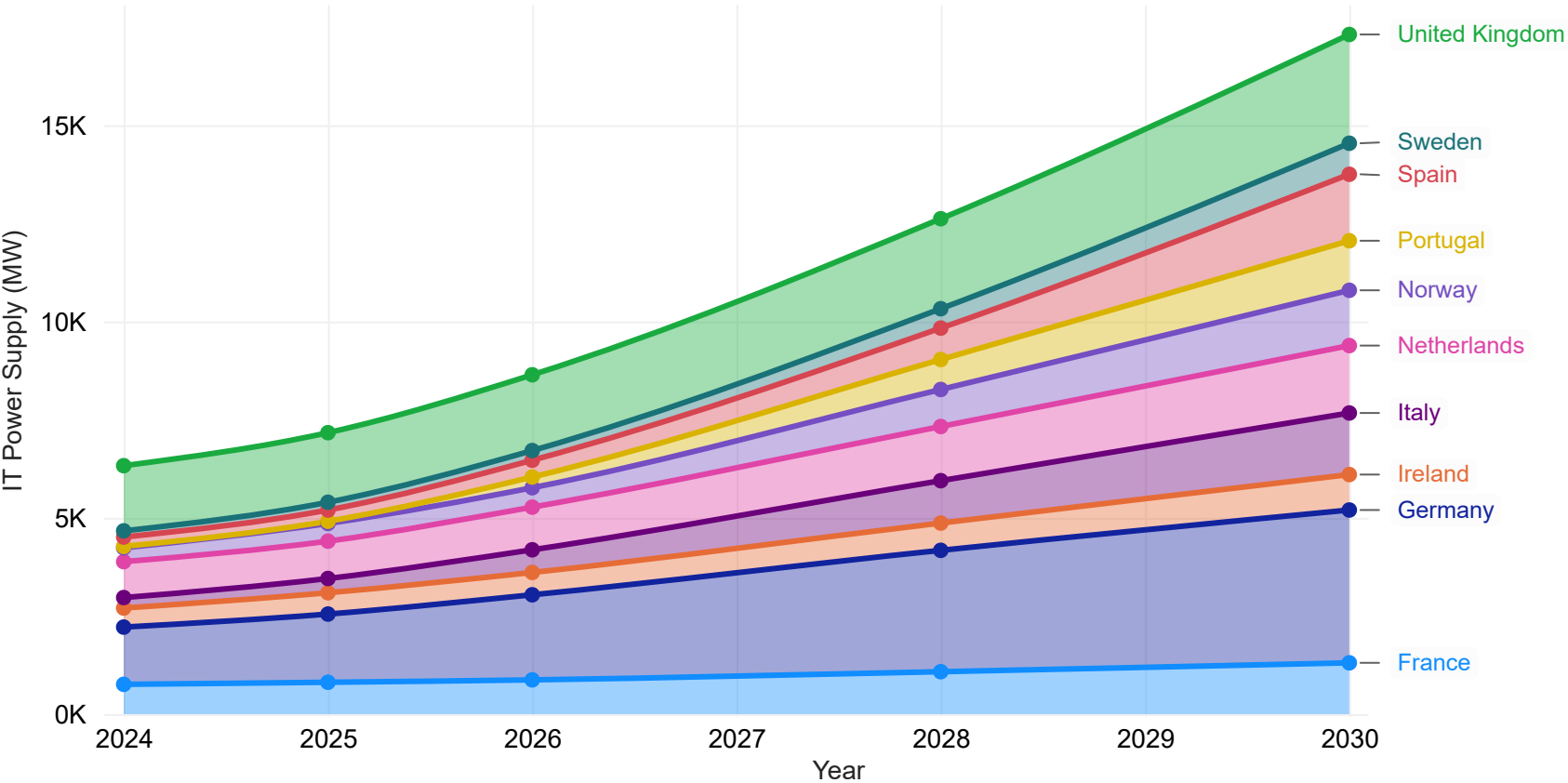
Since I lacked time, I could not deepen some aspects that a rigorous data analyst would have better studied. And I used AI to speed up some necessary tasks.

However, the strategy behind the analysis is only mine.

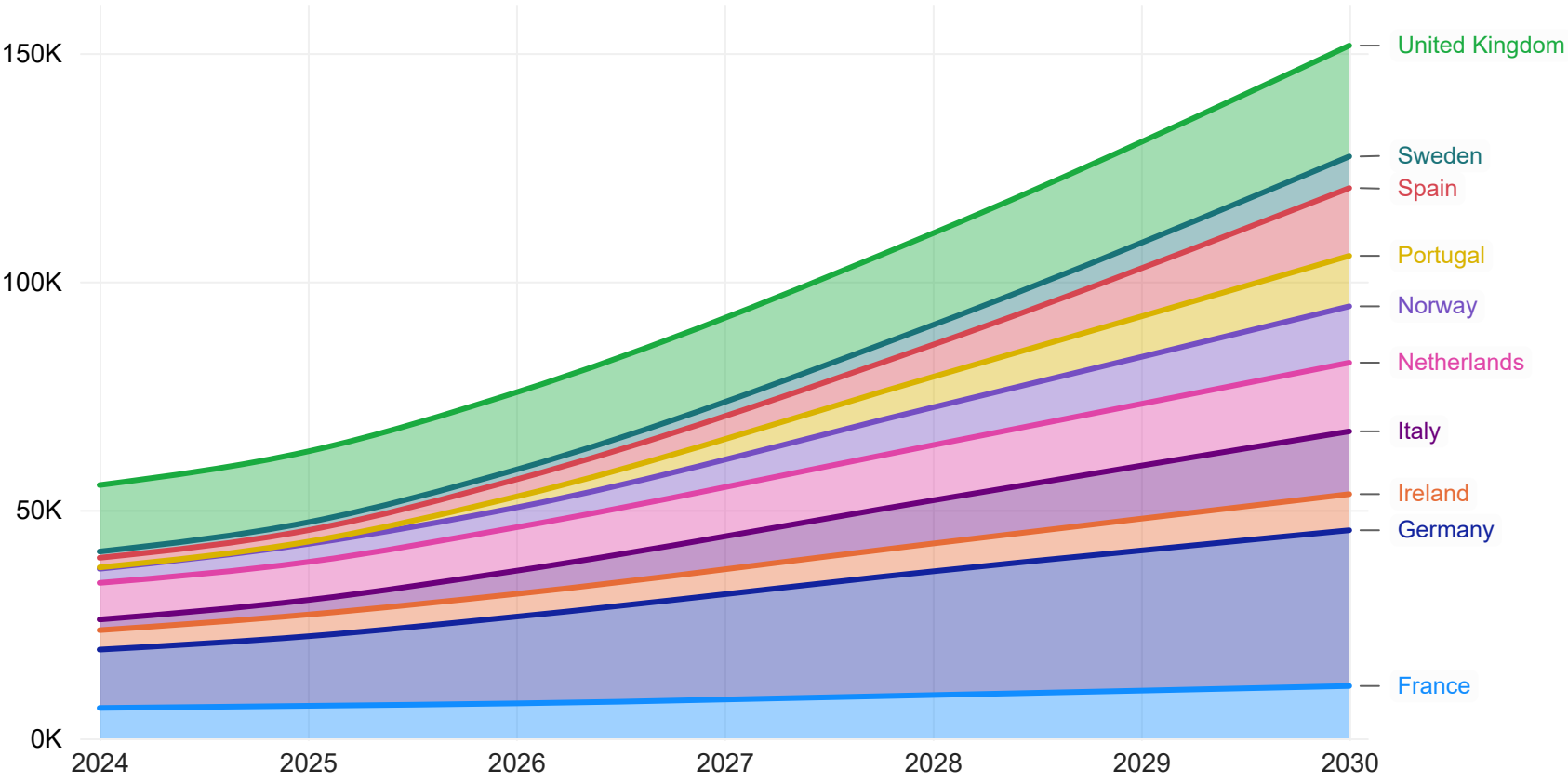
Please, if you'd like to continue my work, visit the related GitHub folder [here](#).

Federica Briganti

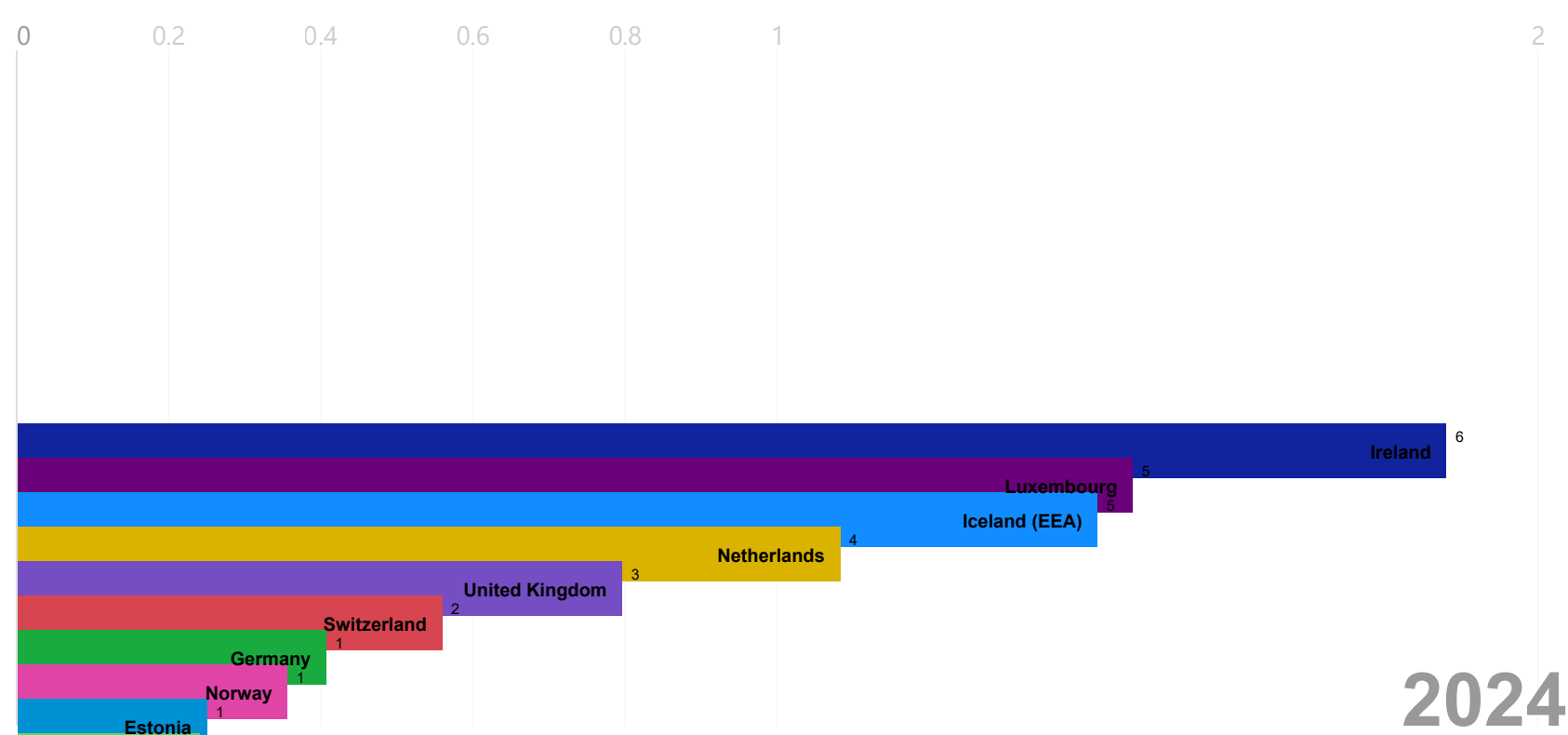
IT Power Supply (MW)
By Year and Country



Yearly Power Consumption (GWh)
By Year and Country



Energy Exposure (%)
By Year and Country



2024

Inputs

IT power Supply (MW)

Source: European Data Centre Association, Pb7 Research. (2026). Europe. (Statista).
Definition: for each European country, the electrical power capacity (MW) dedicated specifically to servers in colocation data centers.

Yearly electricity demand (GWh) for 2028 and 2030

Source: ENTSO-E, European Resource Adequacy Assessment (ERAA).
Definition: projected total national electricity demand (TWh/year) for two specific future "Target Years", 2028 and 2030, as modeled by the European association of electricity transmission system operators for their own grid adequacy planning.

Monthly electricity demand (GWh) from 2015 onwards

Source: Monthly Energy Statistics (IEA).
Definition: measured monthly electricity consumption by country from 2015 onwards.

Output

Energy exposure (%)

Formula: (planned IT power, MW, converted to an annual GWh equivalent) ÷ (national annual electricity consumption, GWh) × 100.
Definition: share of the country's current electricity demand related to planned data centers if they are built and running continuously.
Purpose: a country with a small grid can be more exposed to the same absolute number of megawatts than a country with a much larger grid.

Conclusions

Ireland has the highest exposure of any EU country in every year (14.33% in 2026, rising to 17.58% in 2030). Portugal shows the steepest growth of all (4.34% → 10.61% → 14.74%), overtaking most other high-exposure countries by 2030. Most EU countries stay under 2% throughout. The EU27 aggregate itself more than doubles over the period (1.78% in 2024 to 4.40% in 2030), showing this is a rising trend at the EU level, not only a story about one or two outlier countries.

Digitalization Growth Rates (pp)
By Country, Period 2021-2024

Country	AI Adoption (pp)	Cloud Services Adoption (pp)	Hosting Data Net Turnover (pp)
Albania		10,18	
Ireland	3,93	9,28	
Luxembourg	9,34	8,89	
Montenegro		6,31	
Finland	20,73	3,46	10,58
Greece	-47,05	3,24	106,31
Switzerland			-12,2
Lithuania	5,23	14,65	138,61
Czechia	26	7,27	17,33
Denmark	15,17	4,36	21,24
Cyprus	8,5	1,87	235,97
France	19,98	4,24	25,72
Estonia	25,27	2,07	27,62
Portugal	-4,57	4,74	30,38
Bosnia and Herzegovina		15,97	30,85
Slovenia	10,33	1,43	32,29
Slovakia	-7,39	-0,71	35,95
Italy	12,98	8,03	41,95
Netherlands	8,39	6,08	42,07
Bulgaria		4,87	43,88
Romania	-7,21	7,53	50,06
Sweden	0,75	-3,58	51,95
Croatia	13,86	2,85	52,83
Germany	16,94	8,85	55,37
Iceland			56,59
Latvia	29,78	11,37	57,11
Belgium	18,96	3,7	61,9
Norway	7,48	4,92	65,93
Austria	34,12	8,88	66,45
Malta	18,61	13,82	7,55
Spain	11,09	4,91	79,64
Hungary	-26,3	21,93	81,87
Poland	14,1	26,49	91,61
Total	0,75	-0,71	10,58

Inputs from [Eurostat](#)
Cloud adoption (%)

Definition: the percentage of enterprises in each country that report buying cloud computing services, surveyed roughly every one to two years from 2014 to 2025.

AI adoption (%)

Definition: the percentage of enterprises in each country that report using artificial intelligence technologies for marketing or sales purposes, surveyed in 2021, 2023, 2024 and 2025.

Hosting Data Turnover (million €)

Definition: the annual net turnover, in million euro, of enterprises classified under "Data processing, hosting and related activities; web portals", by country, from 2021 to 2024.

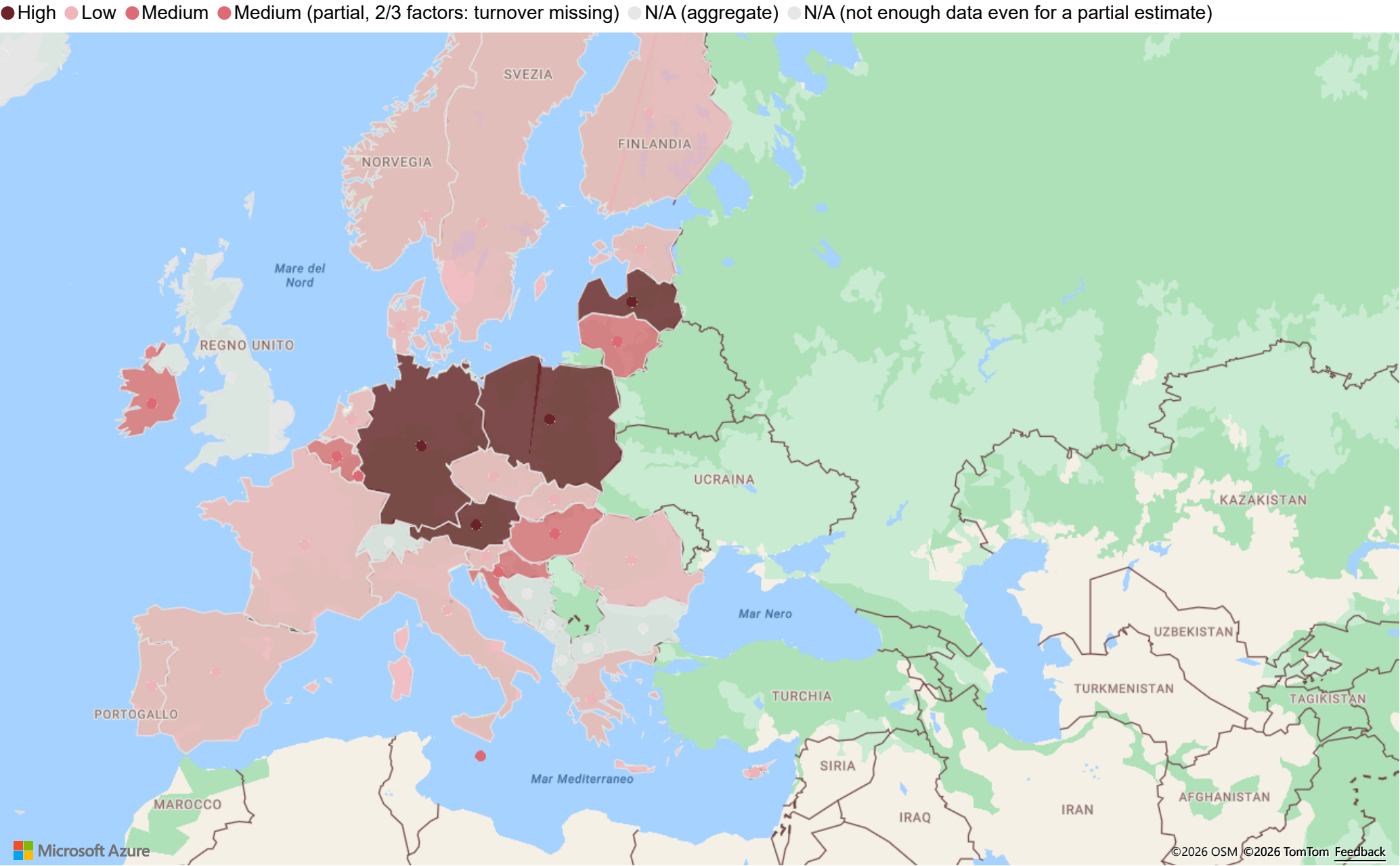
Output
Digitalization Velocity

Definition: for each country, the growth of each of the three factors between 2021 and 2024 is compared to the EU's own growth in the same period. The level is simply how many of the three factors grew faster than the EU: 3 of 3 → level "High". 2 of 3 → level "Medium". 0-1 of 3 → level "Low". If only the turnover figure is missing, a partial KPI is computed using cloud and AI growth only, and is explicitly labeled "(partial, 2/3 factors)" rather than presented as a full result.
Purpose: turns three separate, differently-scaled indicators into one comparable score, so it is possible to say "this country's digital economy is growing faster/slower than the EU average" in a single number.

Conclusions

Germany, Austria, Latvia and Poland are the only countries with a full "High" KPI (growing faster than the EU average on all three factors simultaneously). Most countries (16 of the countries with a full KPI) are "Low". Growth on the three factors does not move together within a country, showing the three factors can tell different stories about the same country, which is the reason for tracking all three rather than one combined figure alone.

Digitalization Velocity
By Country, Period 2021-2024



Anomaly Score Weighted Average (%)
By Country and Year

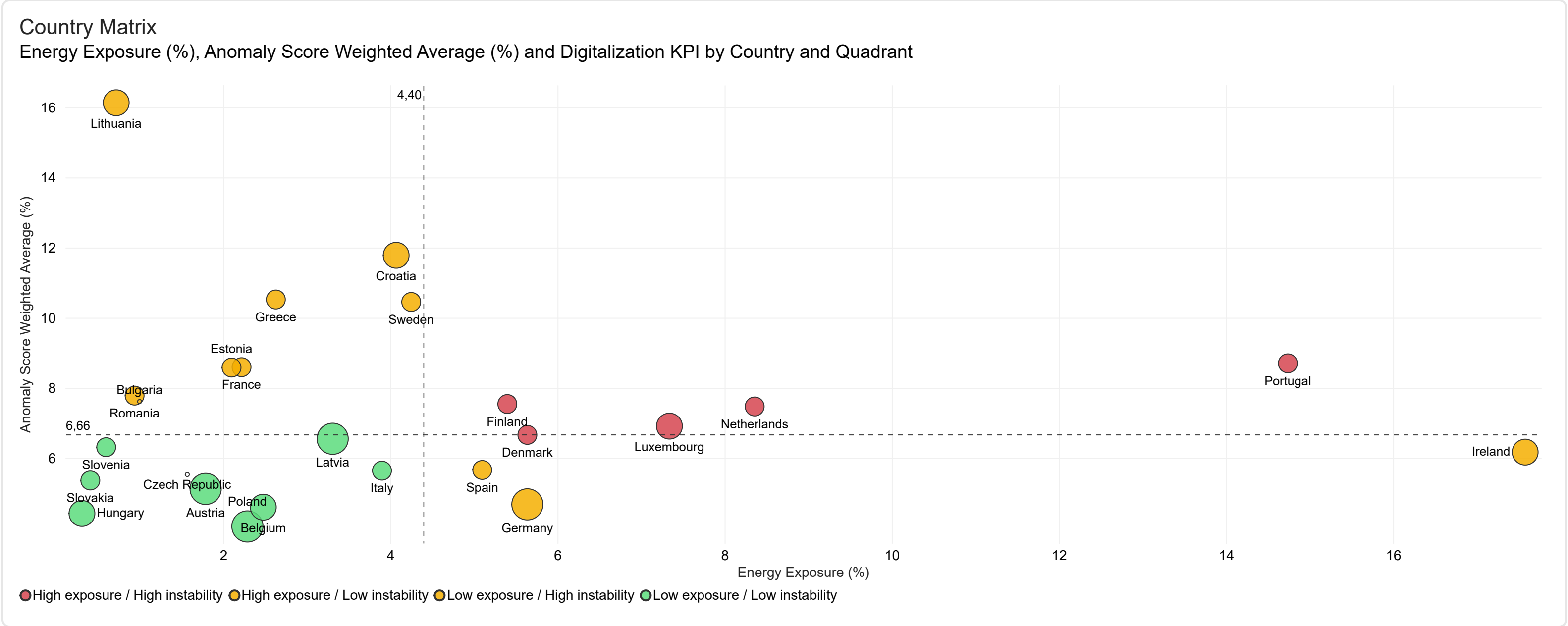
Country	⊕ 2010	⊕ 2011	⊕ 2012	⊕ 2013	⊕ 2014	⊕ 2015	⊕ 2016	⊕ 2017	⊕ 2018	⊕ 2019	⊕ 2020	⊕ 2021	⊕ 2022	⊕ 2023	⊕ 2024	⊕ 2025	Total ▼
Lithuania			13,44				11,08	18,81				11,41			11,64	16,00	16,13
Cyprus							10,70		12,40		12,94	8,96	15,61	8,29	15,90	6,60	12,35
Croatia						21,58	11,33		7,78		7,47	8,06			8,83		11,78
Greece	10,56	7,32		9,03		7,94	10,09	8,33	8,91	8,74	8,67	9,15	8,82	10,08	14,28	9,36	10,52
Sweden	12,24	7,71	8,54				15,01	6,84	8,61		7,68	9,34	7,80	7,34	8,55	8,14	10,45
Malta							11,24	11,08	7,55		10,03	8,48	5,28	8,09	12,14	8,98	9,54
Portugal	7,00	6,48	11,34	12,99	3,82		7,40	5,97	4,94	6,58	8,41	7,56		5,09			8,70
France	15,34	7,01	15,42	8,47		9,13	7,40	9,96	7,78		7,97	6,89	7,62	5,77	7,95		8,59
Estonia	10,00	6,08	15,02	5,03	6,91	6,91	7,38	8,44	9,69	7,21	6,40	13,82	8,04	7,78	7,26		8,58
Romania						6,37	7,47	6,44	4,68	5,61	7,52	5,75		5,71	7,47	10,87	7,78
Bulgaria						5,45	5,44	8,73	7,61	7,71		6,36	10,34	5,97	5,99	9,62	7,61
Finland	8,28	6,84	6,55	5,76	4,58	5,71	9,54		6,59	4,09	7,53	11,17	4,93	6,15	8,47		7,54
Netherlands					3,08		9,50	4,81	8,53	4,68	5,63		6,11				7,47
Luxembourg	4,00	5,29	8,39	5,40				3,86			8,19		4,96	4,19			6,91
Denmark	9,08	5,37	7,46		5,45						9,67	5,75	6,68	7,03	4,82	6,24	6,66
Latvia	7,08	1,41	10,78	8,79	3,87		3,81					5,27	5,25		3,90	4,89	6,55
Slovenia	6,71	4,92	5,31	4,73					4,20	5,37	6,33		5,22	3,05	6,46	8,01	6,31
Ireland	9,43	3,34	5,34	4,18			5,07	3,06	8,40	6,97	5,93	4,51		4,77	3,68		6,17
Spain	4,78	4,60	7,48	5,30		5,19	4,94	4,46		2,67	7,28	4,41	5,43	2,11	5,08	5,76	5,66
Italy	2,54	3,95	4,31		5,20	5,02	3,62	3,30			6,88		4,62		5,25	4,89	5,64
Czech Republic	8,83	1,55	9,49	3,80	5,26	5,59		5,36	5,62		7,44		4,18	3,91	4,60		5,53
Slovak Republic		4,43	4,73	4,87	6,01			6,25	4,13	4,85	6,21	3,41	5,34	4,87	4,75	5,50	5,36
Austria	4,61	3,61	8,49	4,47	4,63	0,76	2,63	3,38	4,11	4,43	6,30	3,06		4,60		4,51	5,12
Germany	8,08	3,32	6,57	3,93	3,46	3,90	2,92	5,91	3,83	3,84	5,54		4,79		2,60		4,68
Belgium	7,17	1,40	7,49	4,34		4,22		2,84	2,85		7,34	3,26	4,79	4,13	3,41	3,99	4,60
Hungary			4,63		3,67		4,72	3,68			4,71	3,60	3,61	4,69	4,89		4,43
Poland	6,96	0,82	5,35	3,32	3,51	3,22		3,85	0,72	3,77	4,68	2,37	3,53	5,05	2,94	3,38	4,05
EU27						2,50		1,73	3,07	2,25	5,14	2,82	3,77	3,58	2,66		3,67

Inputs
Monthly electricity demand (GWh) from 2015 onwards
See Step1.

Output
Anomaly Score Weighted Average

Formula: each country's monthly series is deseasonalized using a classic additive decomposition (a 12-month centered moving average for the trend, average deviation by calendar month for the seasonal component), producing a residual that isolates whatever is not explained by trend or season. Isolation Forest is then run on that residual to flag anomalous months, decided by the model separately for each country. Each flagged anomaly then gets a weight built from two components multiplied together: a recency component that decays exponentially with age (halving every 5 years), and a severity component equal to the size of the anomaly itself.
Definition: a this combined-weight average size of a country's anomalies.
Purpose: acts as a proxy for how much resilience a national grid already has. A country whose consumption has recently shown large, hard-to-explain swings has, by this measure, less slack to absorb new load such as data center demand without further instability.

Conclusions
the share of anomalous months varies by country (from 10.8% in Czechia to 21.6% in Slovakia). Lithuania stands out with by far the largest combined-weight average anomaly size (16.13%, well above the next highest countries), while the EU27 aggregate's own anomalies are much smaller (3.67%) than any individual country's, expected, since country-level swings partly cancel out once summed at EU level.



Inputs
Energy Exposure per country/year (Step 1), Digitalization Velocity per country (Step 2), and Anomaly Score Weighted Average per country (Step 3).

Output
Each country/year is classified on two axes. On the exposure axis, a country is "high" if its Exposure_pct (Step 1) is at or above that year's EU27 aggregate exposure. On the instability axis, a country is "high" if its Anomaly Score Weighted Average (Step 3) is at or above the median value across all countries. The median is used here instead of the EU aggregate, because the EU aggregate consumption series is structurally smoother, which would make almost every individual country look "unstable" by comparison. Digitalization Velocity (Step 2) is layered onto the same chart as point size.

- Conclusions**
- **Portugal** needs the most attention: the steepest exposure growth of any country in the sample, stays in the High exposure/High instability quadrant every year, and has a Low digitalization velocity, Its infrastructure growth is not matched by a comparable growth of its domestic digital economy. The mismatch might reflect data centers built to serve international demand rather than the country's own digital economy.
 - **Ireland and Luxembourg** have the highest exposure in the sample but stay in the High exposure/Low instability quadrant every year. High exposure has not, so far, come with grid instability.
 - **Germany** is the most balanced profile with high digitalization velocity, moderate and steadily growing exposure, low consumption instability.
 - **The Netherlands, Denmark and Finland** enter or settle into the High exposure/High instability quadrant as the EU exposure threshold itself rises over time.